

Title: Pedigree-Based Investigation of a Familial Neurobehavioral Disorder

Student name: FATEMEH YAGHMOORI

Affiliated with:

- **Comprehensive Genomics Research Laboratory (Iran)**
- **Helix Institute for Medical & Biological Sciences (USA)**

- **Clinical Features:**

The family has two children, an affected daughter and five healthy daughters, through consanguineous marriage (cousin and cousin). Family history is also seen in the pedigree. The uncle of the affected children also has similar symptoms and has an affected son and a healthy daughter and son. The patients are unteachable, aggressive, hyperactive, restless, delayed and unbalanced in walking. The probands were 37 and 28 years old at the time of examination, with head circumferences of 57 (+2.42 SD) and 56 (+1.52 SD), respectively.

- **Methodology:**

(Bioinformatic Analysis and Variant Filtering Pipeline)

To identify potential causal variants associated with the observed phenotype, a stringent bioinformatic filtering pipeline was implemented on the Whole Exome Sequencing (WES) data. The filtering strategy was executed as follows:

- 1- **Functional Annotation:** To focus on high-impact alterations, variants were restricted to functional regions, specifically prioritizing exonic and splicing alterations (RefSeq: exonic/exonic, splicing) within the refGene annotation.
- 2- **Pathogenicity Prediction:** Variants were prioritized based on their predicted functional impact using automated InterVar analysis. The filtering criteria included only variants classified as Likely Pathogenic (LP) or of Uncertain Significance (VUS), ensuring that benign variants were excluded from the primary analysis.
- 3- **Genotype Filtering:** To capture both dominant and recessive inheritance models, the analysis was restricted to heterozygous (0/1) and homozygous (1/1) genotypes.
- 4- **Population Frequency Filtering:** To exclude common polymorphisms, a strict population frequency threshold was applied. Only rare variants with a minor allele frequency (MAF) of ≤ 0.00 (absent from large-scale population databases) or specific frequency constraints (e.g., up to 0.50 for certain genetic models) were retained for further investigation.

5- Candidate Gene Selection (**NAGA**): Integrating the clinical phenotype with the bioinformatic filtration highlighted the NAGA heterozygous stop-gain variant (c.892C>T; p.Arg298Ter*) as the primary candidate. This variant introduces a premature termination codon predicted to result in a truncated, non-functional protein or nonsense-mediated decay (NMD), perfectly matching the lysosomal enzyme deficiency spectrum and neurological presentation observed in the probands.

The integrated approach ensured a focused downstream analysis on rare, high-impact candidates that are most likely to contribute to the patient’s clinical presentation.

❖ **Validation of Variant Significance and Functional Impact**

Following the initial filtering pipeline, candidate variants classified as Likely Pathogenic (LP) or Uncertain Significance (VUS) underwent further interrogation using established databases and classification tools to validate their potential pathogenicity:

- 1- ClinVar Database Cross-referencing: Candidate variants were systematically cross-referenced against the ClinVar database to assess their reported clinical significance, including pathogenicity assertions (Pathogenic, Likely Pathogenic, Benign, Likely Benign, VUS) and associated clinical phenotypes. This step aimed to leverage existing expert interpretations and published evidence.
- 2- VarSome Analysis for ACMG Classification: For a comprehensive assessment according to the ACMG/AMP guidelines, candidate variants were analyzed using the VarSome platform.

Genomic Nomenclature (GRCh37)	Gene/Transcript	Variant Nomenclature	Zygosity	Phenotype/Inheritance	Molecular consequence	Classification
Chr22: 42062892 C>T	NAGA/ NM_000262.3	c.892C>T, p.Gln298Ter	Het	Schindler disease, type I /AR Schindler disease, type III /AR	nonsense	Likely Pathogenic/ Uncertain significance (VUS) (Based on ClinVar and VarSome)

- **Clinical Interpretation of Nonsense Variant in NAGA:**

- 1. Variant Details:
Gene: NAGA (NM_000262.3)

Variant: c.892C>T (p.Gln298Ter)

Zygosity: Heterozygous (0/1)

Classification: Likely Pathogenic (based on LoF mechanism)

Comment: This sequence change creates a premature translational stop signal (p.Gln298) in the NAGA gene. It is expected to result in an absent or disrupted protein product. Loss-of-function variants in NAGA are known to be pathogenic (PMID: 8782044, 11251574). This variant is present in population databases (no rsID available, gnomAD 0.003%). This variant has not been reported in the literature in individuals affected with NAGA-related conditions. For these reasons, this variant has been classified as Pathogenic.

- **Description:**

Alpha-N-acetylgalactosaminidase (NAGA) deficiency is a very rare lysosomal storage disorder. It is clinically heterogeneous with 3 main phenotypes: type I is an infantile-onset neuroaxonal dystrophy; type II, also known as Kanzaki disease, is an adult-onset disorder characterized by angiokeratoma corporis diffusum and mild intellectual impairment; and type III is an intermediate disorder with mild to moderate neurologic manifestations (Desnick and Schindler, 2001).

2. Molecular Impact:

The variant *c.892C>T* introduces a premature termination codon at amino acid position 298 (p.Gln298Ter). This is a nonsense mutation that is highly predicted to result in a truncated protein or undergo Nonsense-Mediated Decay (NMD), leading to a complete loss of enzymatic function (Loss-of-Function).

3. Clinical Correlation:

Deficiency in the NAGA gene is associated with Mucopolysaccharidosis II and III. While the hallmark of these lysosomal storage disorders typically involves skeletal abnormalities and growth retardation, the neurodevelopmental impact (including cognitive impairment and neurological decline) is a well-documented feature. Pathogenic variants in this gene are known to cause a spectrum of disorders, ranging from the severe neuroaxonal dystrophy known as Schindler disease (Type I) to the milder phenotype known as Kanzaki disease (Type II/III).

4. Recommendation:

Biochemical Confirmation: It is strongly recommended to confirm the genetic finding by measuring α -N-acetylgalactosaminidase enzyme activity in leukocytes or fibroblasts.

Urine Oligosaccharide Analysis: Testing for the presence of specific glycopeptides/oligosaccharides in urine is a standard diagnostic step for Schindler/Kanzaki disease.

Clinical Review: Correlation with other signs of lysosomal storage, such as angiokeratoma (skin lesions) or ophthalmic findings, is advised.

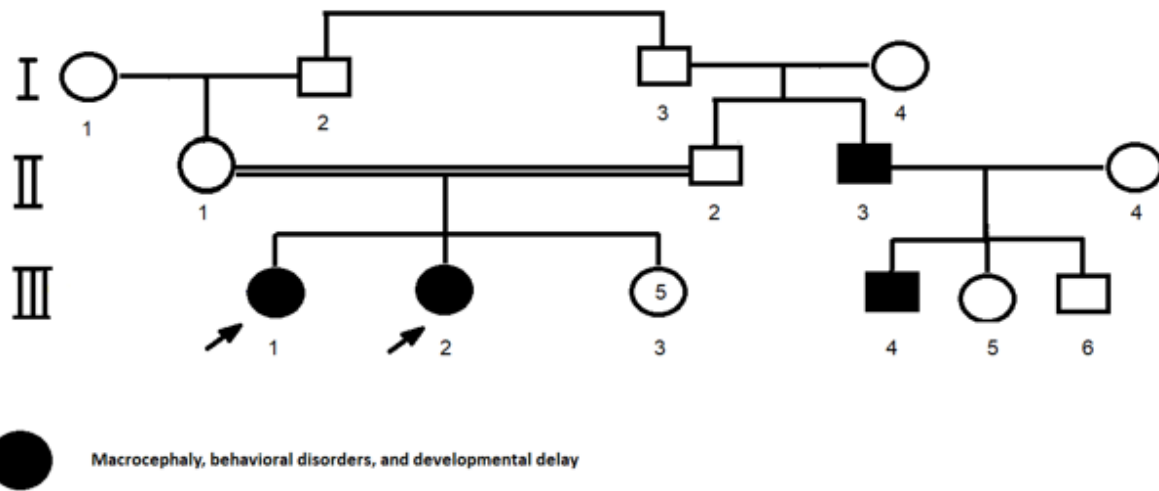
Clinical Correlation and Gene Interpretation:

The Whole Exome Sequencing (WES) analysis identified a candidate pathogenic variant in the NAGA gene (c.892C>T), which is a stop-gain mutation resulting in a predicted truncated protein. The identified phenotype of the probands—characterized by macrocephaly, neurological impairments, and ataxia—demonstrates a significant clinical correlation with the known spectrum of $\alpha\alpha$ -N-acetylgalactosaminidase deficiency.

According to established literature, specifically the comprehensive study by Bakker et al. (2001), deficiency of the $\alpha\alpha$ -NAGA enzyme leads to a heterogeneous group of lysosomal storage disorders (including Schindler and Kanzaki diseases). A critical aspect of this disorder is its profound genotype-phenotype discrepancy. The literature highlights that patients exhibiting various neurological manifestations, such as intellectual disability and motor dysfunction, can arise from diverse mutations, and the severity of the disease does not always follow a linear correlation with the type of genetic lesion.

In the present case, the identification of the c.892C>T variant in NAGA provides a strong molecular explanation for the observed neurological and developmental symptoms. The presence of ataxia and neurological decline in the probands aligns with the lysosomal storage mechanisms described in the context of $\alpha\alpha$ -NAGA deficiency, where the failure to catabolize specific glycoconjugates leads to cellular dysfunction in neural tissues. Therefore, based on the clinical phenotype and the highly relevant literature, this variant is prioritized as a primary candidate for the underlying genetic etiology.

- **Pedigree:**



- **References:**

1. Bakker, H. D., de Sonnaville, M. L. C. S., Vreken, P., Abeling, N. G. G. M., Groener, J. E. M., Keulemans, J. L. M., & van Diggelen, O. P. (2001). Human α -N-acetylgalactosaminidase (α -NAGA) deficiency: no association with neuroaxonal dystrophy?. **European Journal of Human Genetics**
2. Richards, S., Aziz, N., Bale, S., Bick, D., Das, S., Gastier-Foster, J., ... & ACMG Laboratory Quality Assurance Committee. (2015). Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology.